

The Machine Age

Machine is a device which replaces or simplifies human or animal's efforts to achieve any physical task. Remember, this is a very simple description of a machine. Simple machines are lever, wedge, wheel, pulley and screw. Man of every age has to overcome obstacles of every kind to complete his task. His source of energy may have fallen short of the requirement; therefore, he has to use some devices to meet his needs. Hence, man invented different machines out of the resources available to him at that point of time.

Out of many machines invented by man, you already know different machines which have proved land mark and took the society forward.

When only wood, stone and bones were available, man created tools to fulfill his needs. Hammer, axe and spears or any other tool were machines. Wheel, one of the greatest inventions of that time, is also a machine.



The early ships used wind power.
Ships fitted with steam engines move without sails

When he discovered copper, he put it to many uses such as making utensils and heads of arrows and spears. Utensils are not machines but heads of arrows, spears and axes. Later, he found iron and in many cases he replaced use of copper with that of iron.

At that stage, the source of energy was either man power or animal power. In sailing boats wind power was also being used along with man power.

It took many years to replace these sources of energy. In the year 1769, James Watt used steam power to run water pumps. You know rest of the story how steam power was used to run many types of machines including the railway engine and ships.

Another point to be noticed is that there was a shift in the source of energy from man and animal power to steam power. Steam could be generated by burning either wood or coal. Wood was to be obtained from forests and coal from mines. At the time of burning both produced carbon-di-oxide, other gases and ash. At that time people hardly noticed about the effect of carbon-di-oxide on the environment. Also that wood and coal are exhaustible.

By resolving one problem man has created another problem.

Petroleum, which is known as crude oil was known to ancient people through surface seepage. It was used by them to build roads, to burn as oil in lamps and to paint their ships. Petroleum is formed from water plants and animals which lived and died hundreds of millions years ago. Their remains got changed into petroleum through natural forces. Today it replaces coal as the primary fuel in industrial and domestic heating. Some of the products coming out of petroleum are petrol, diesel, kerosene and many chemicals which are used to manufacture solvents, plastics, synthetic rubber, fiber, soaps, paints waxes, jellies, explosives and fertilizers. Petroleum is also exhaustible.

Again, discovery of petroleum gave tremendous boost to man's creativity. Different machines were invented around this source of energy. Automobile, aeroplane, diesel run engines powering automobiles, railways and ships started serving the society. All this happened between 18th and 20th century. This also gave boost to generation of electric power in places where hydroelectricity was not possible. Now electricity is being produced by burning coal or diesel or using water. Like coal as a source of energy, petroleum generates carbon-di-oxide and carbon monoxide gases. These are called green house gases. These gases are creating problems to mankind which are life threatening.

Meanwhile, another source of energy came to the fore front in late 20th century. It is the Atomic energy. Presently it is used to generate electrical energy.

ALTERNATIVE SOURCES OF ENERGY

This shift in the sources of energy, from animal energy to coal energy to petroleum energy and lastly to atomic energy had been a shift in man's approach to his needs. But still the major energies used are steam, gas, petrol, diesel, electricity. In some cases animal energy is still used. These have created more machines serving the humanity. Now man is seriously working on clean sources of energy.



Spinning machines twist the cotton into long threads

These alternative sources of energy presently available are wind-energy, solar-energy and hydrological-energy.

In Wind-energy, force of wind is used to run windmill which in turn may run a machine directly or run a dynamo to produce electricity. This electricity charged batteries which can be used as and when needed.



Alexander Volta made the first cell to generate electric current

In Solar-energy, the sun's energy can be used to cook food through solar cooker, heat water or produce electricity and light solar lamps.

Wind-energy and solar-energy are not cost effective as on today.

Hydro-energy is not available everywhere.

Things to Remember

- Machine is a device which replaces or simplifies human or animal's efforts to achieve any physical task,
- Out of many machines invented by man, there are some machines which have proved landmark and took the society forward,
- Steam is generated by burning either wood or coal. Wood is obtained from forests and coal from mines. Both are exhaustible and generate green house gases,
- Petroleum is formed from water plants and animals which lived and died hundreds of millions years ago. Their remains got changed into petroleum through natural forces, Today petroleum replaces coal as the primary fuel in industrial and domestic heating.
- Some of the products coming out of petroleum are petrol, diesel, kerosene and many chemicals which are used to manufacture solvents, plastics, synthetic rubber, fiber, soaps, paints, waxes, jellies, explosives and fertilizers. This is also exhaustible. It also generates green house gases,
- Another source of energy came to the fore front in late 20th century is the Atomic energy. Presently it is used to generate electrical energy,
- The major energies used are steam, gas, petrol, diesel, electricity,
- The alternative sources of energy presently available are wind-energy, solar-energy and hydrological-energy. These are called clean sources of energy. But, as on today, these are not cost effective.